

Automation Level 3 Badge

Participant Resource Sheet

Name:_____

Email used for badges: _____

Topic	Task	Mentor Signature	Date
Baseline and Diagnostic (Start)	Take the Badge 3 diagnostic, trace and change Cobra's baseline in the Badge3Starter fork, start the migrated library, and record the work plan; complete the independent mastery attempt after Step 14. Step 1		
Calculus for Controls	Correct the calculus worksheet to full credit and apply rates and accumulation to a saved Badge 2 elevator log. Step 2		
Advanced Output Capabilities	Add typed torque-current requests that refuse unsupported or nonfinite values visibly; trace REAL, SIM and REPLAY construction with a replay and a SIM construction test. Step 3		
Roller Sensing and Jams	Compare Talon FX voltage, FX torque-current and FXS voltage roller effort, add CANrange presence with a physical test, and latch a sustained jam until reset. Step 4		
Flywheel Characterization	Compare flywheel speed control across output modes, adapt Cobra's quasistatic characterization for the flywheel, feeder and intake roller with bounded cleanup, and validate the gains on a separate run. Step 5		
FuelSim Launch Experiment	Pin FuelSim, model launch velocity from measured flywheel speed, radius and slip, work out the launch elevation from the hub table, and show a scoring shot and a controlled miss. Step 6		
Pivot Reference and Profiles	Refine Motion Magic on FX and FXS, establish manual and absolute references, bound travel, characterize the pivot, show physical repeatability, and complete the Expo comparison and worksheet. Step 7		
Logged Mechanism and CAD	Build the measured and goal intake pivot drawing, verify known poses on it and the animated CAD, and		

	review a saved log and replay with another member. Step 8		
Subsystem-Owned States	Integrate roller, flywheel and pivot states and test refused transitions, readiness failure, timeout, interruption and recovery. Step 9		
Object Detection	Build object-detection IO, locate a stationary piece from camera rays with stale-frame handling, and check a physical camera at known locations. Step 10		
Combined Simulation	Combine maple-sim, FuelSim and PhotonVision in one simulation, run open-space and contact BLine experiments with the true and estimated pose logged separately, and have another member reproduce them. Step 11		
One-Piece Autonomous	Collect one piece and record a simulated score with BLine, camera observations and owned states, and prove the no-piece, target-lost, timeout, never-ready and interruption cases. Step 12		
Hardware Evidence	Verify the FX physical voltage baseline, demonstrate torque-current roller, flywheel and pivot with one voltage and torque-current comparison, review the Badge 2 controller route, and record handoff data. Step 13		
Diagnosis, Review, and Teaching	Diagnose prepared and personal logs, lead a code review, help a Badge 1 or 2 student, and propose a lesson improvement. Step 14		
Final Check, Handoff, and Sign-Off	Score 100% on the independent final, hand off the project for another member to build and reproduce, and receive the mentor's final review. Step 15		